

Solution of P3P Problem

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These slides follow Appendix A in the paper

Fischler-Bolles: Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography, 1981.

Recall, the camera projection of n points is given by

$$\lambda_i \mathbf{u}_i = \mathbf{M} \mathbf{X}_i, \quad i = 1, \dots, n \quad (1)$$

- ◆ $\mathbf{u}_i$ is image point (homogeneous, i.e. 3-vector) measured in the CCD coordinate frame
 - ◆ $\mathbf{X}_i$ is scene point (homogeneous, i.e. 4-vector) measured in the scene coordinate system
 - ◆ $\mathbf{M} = \mathbf{K} \mathbf{R} [\mathbf{I} \mid -\mathbf{T}]$ is camera projection matrix, where
 - $\mathbf{K}$ is (upper triangular) matrix of intrinsic parameters
 - $\mathbf{R}$ is rotation matrix (i.e., $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ and $\det \mathbf{R} = 1$)
 - $\mathbf{T}$ is translation (nonhomogeneous, i.e. a 3-vector). Note, camera center is just $\mathbf{T}$ in homogeneous coordinates, because $\mathbf{M} \begin{bmatrix} \mathbf{T} \\ 1 \end{bmatrix} = \mathbf{0}$.
 - $\mathbf{I}$ denotes the 3×3 identity matrix
- $\mathbf{K}$ are the **intrinsic** (inner) and $\mathbf{R}, \mathbf{T}$ are the **extrinsic** (outer) camera parameters.
- ◆ Full camera calibration: Find $\mathbf{K}, \mathbf{R}, \mathbf{T}$ from corresponding pairs $\{(\mathbf{u}_i, \mathbf{X}_i)\}_{i=1}^n$
 - Easy: By solving equation (1), which is a homogeneous linear system for $\mathbf{M}, \lambda_i$
 - We need at least $n = 6$ correspondences
 - ◆ Calibration of extrinsics only: Find $\mathbf{R}, \mathbf{T}$ from matrix $\mathbf{K}$ and pairs $\{(\mathbf{u}_i, \mathbf{X}_i)\}_{i=1}^n$
 - More difficult, because (1) becomes a system of non-linear equations for $\mathbf{R}, \mathbf{T}, \lambda_i$
 - We need at least $n = 3$ correspondences, which leads to P3P problem.

P3P problem: Given $\mathbf{K}$ and three corresponding pairs $\{(\mathbf{u}_i, \mathbf{X}_i)\}_{i=1}^3$, find $\mathbf{R}$ and $\mathbf{T}$.

- ◆ That means, solve equation system

$$\lambda_i \mathbf{u}_i = \mathbf{K} \mathbf{R} [\mathbf{I} \mid -\mathbf{T}] \mathbf{X}_i, \quad i = 1, \dots, 3 \quad (2)$$

where the knowns are $\mathbf{K}, \mathbf{u}_i, \mathbf{X}_i$ and the unknowns are $\mathbf{R}, \mathbf{T}, \lambda_i$.

- ◆ Denote $\mathbf{v}_i = \mathbf{K}^{-1} \mathbf{u}_i$. Let $\mathbf{x}_i$ denote *non-homogeneous* points $\mathbf{X}_i$, i.e., $\mathbf{X}_i = \begin{bmatrix} \mathbf{x}_i \\ 1 \end{bmatrix}$.

Now, we obtain the final **P3P equation system**

$$\lambda_i \mathbf{v}_i = \mathbf{R}(\mathbf{x}_i - \mathbf{T}), \quad i = 1, \dots, 3 \quad (3)$$

where the knowns are $\mathbf{v}_i, \mathbf{x}_i$ and the unknowns are $\mathbf{R}, \mathbf{T}, \lambda_i$, for $i = 1, \dots, 3$. All vectors $\mathbf{v}_i, \mathbf{x}_i, \mathbf{T}$ are non-homogeneous here (i.e., 3-vectors).

- ◆ Eliminate $\mathbf{R}$ by taking the 2-norm (i.e., length) of equation (3):

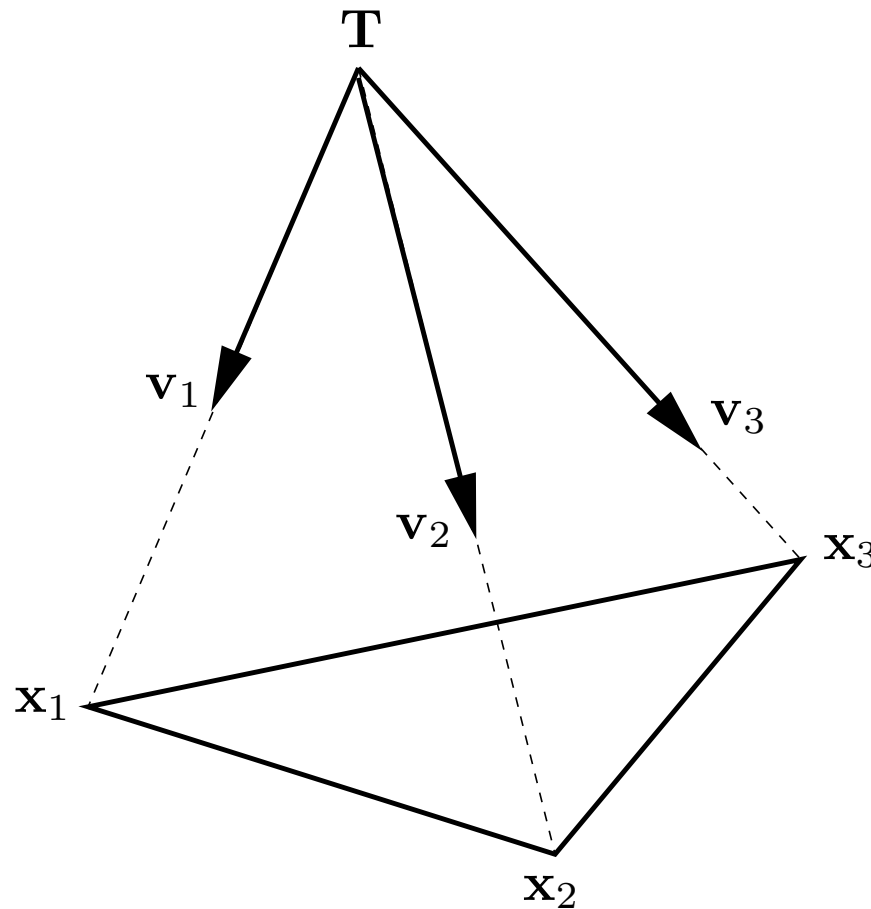
$$|\lambda_i| \|\mathbf{v}_i\| = \|\mathbf{x}_i - \mathbf{T}\|, \quad i = 1, \dots, 3 \quad (4)$$

which holds because any vector $\mathbf{x}$ satisfies $\|\mathbf{R}\mathbf{x}\| = \|\mathbf{x}\|$ (rotation preserves length) and $\|\lambda\mathbf{x}\| = |\lambda| \|\mathbf{x}\|$.

- ◆ Lengths of vectors $\mathbf{v}_i$ play no role because they are multiplied by unknowns λ_i . Therefore, consider only angles between them.

◆ Given:

- Angle cosines: $\cos(\angle \mathbf{v}_1 \mathbf{v}_2) = C_{ab}$, $\cos(\angle \mathbf{v}_1 \mathbf{v}_3) = C_{ac}$, $\cos(\angle \mathbf{v}_2 \mathbf{v}_3) = C_{bc}$
- Distances $\|\mathbf{x}_1 - \mathbf{x}_2\| = R_{ab}$, $\|\mathbf{x}_1 - \mathbf{x}_3\| = R_{ac}$, $\|\mathbf{x}_2 - \mathbf{x}_3\| = R_{bc}$

◆ Wanted: Distances $\|\mathbf{x}_1 - \mathbf{T}\| = a$, $\|\mathbf{x}_2 - \mathbf{T}\| = b$, $\|\mathbf{x}_3 - \mathbf{T}\| = c$ ◆ Geometric meaning: Find the position (rotation+translation) of the (rigidly linked) triplet $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ in space such that the straight lines given by vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ pass through points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$.

The problem leads to a system of 3 quadratic equations with three unknowns as follows.

The law of cosines applied to triangles $\mathbf{x}_1\mathbf{T}\mathbf{x}_2$, $\mathbf{x}_1\mathbf{T}\mathbf{x}_3$, $\mathbf{x}_2\mathbf{T}\mathbf{x}_3$ gives:

$$R_{ab}^2 = a^2 + b^2 - 2abC_{ab} \quad (5a)$$

$$R_{ac}^2 = a^2 + c^2 - 2acC_{ac} \quad (5b)$$

$$R_{bc}^2 = b^2 + c^2 - 2bcC_{bc} \quad (5c)$$

We will eliminate a . Introduce new variables x, y such that $b = xa$ and $c = ya$.

$$R_{ab}^2 = a^2 + x^2a^2 - 2a^2x C_{ab} \quad (6a)$$

$$R_{ac}^2 = a^2 + y^2a^2 - 2a^2y C_{ac} \quad (6b)$$

$$R_{bc}^2 = x^2a^2 + y^2a^2 - 2a^2xyC_{bc} \quad (6c)$$

From equations (6a) and (6c), and from equations (6b) and (6c) we have respectively

$$R_{bc}^2(1 + x^2 - 2xC_{ab}) = R_{ab}^2(x^2 + y^2 - 2xyC_{bc}) \quad (7a)$$

$$R_{bc}^2(1 + y^2 - 2yC_{ac}) = R_{ac}^2(x^2 + y^2 - 2xyC_{bc}) \quad (7b)$$

Denote $K_1 = (R_{bc}/R_{ac})^2$ and $K_2 = (R_{bc}/R_{ab})^2$.

$$0 = (1 - K_1)y^2 + 2(K_1C_{ac} - xC_{bc})y + (x^2 - K_1) \quad (8a)$$

$$0 = y^2 + 2(-xC_{bc})y + [x^2(1 - K_2) + 2xK_2C_{ab} - K_2] \quad (8b)$$

The last equation pair can be written as

$$0 = m y^2 + p y + q \quad (9a)$$

$$0 = m' y^2 + p' y + q' \quad (9b)$$

where m, p, q, m', p', q' are polynomials in x .

Eliminate the terms with y^2 as follows:

- ◆ Multiply (9a) by m' , multiply (9b) by m and subtract. This gives (10a).
- ◆ Multiply (9a) by q' , multiply (9b) by q , subtract, and divide by y . This gives (10b).

$$0 = (pm' - p'm)y + (m'q - mq') \quad (10a)$$

$$0 = (m'q - mq')y + (p'q - pq') \quad (10b)$$

We assumed that $m'q \neq mq'$. Eliminating y gives

$$0 = (m'q - mq')^2 - (pm' - p'm)(p'q - pq') \quad (11)$$

After regrouping terms, we obtain that this is a quartic polynomial in x ,

$$0 = G_4x^4 + G_3x^3 + G_2x^2 + G_1x + G_0 \quad (12)$$

where

$$G_4 = (K_1K_2 - K_1 - K_2)^2 - 4K_1K_2C_{bc}^2$$

$$G_3 = 4(K_1K_2 - K_1 - K_2)K_2(1 - K_1)C_{ab} + 4K_1C_{bc}[(K_1K_2 - K_1 + K_2)C_{ac} + 2K_2C_{ab}C_{bc}]$$

$$G_2 = [2K_2(1 - K_1)C_{ab}]^2 + 2(K_1K_2 - K_1 - K_2)(K_1K_2 + K_1 - K_2) + 4K_1[(K_1 - K_2)C_{bc}^2 + K_1(1 - K_2)C_{ac}^2 - 2(1 + K_1)K_2C_{ab}C_{ac}C_{bc}]$$

$$G_1 = 4(K_1K_2 + K_1 - K_2)K_2(1 - K_1)C_{ab} + 4K_1[(K_1K_2 - K_1 + K_2)C_{ac}C_{bc} + 2K_1K_2C_{ab}C_{ac}^2]$$

$$G_0 = (K_1K_2 + K_1 - K_2)^2 - 4K_1^2K_2C_{ac}^2$$

1. Compute $G_0, \dots, G_4$ from the intrinsic camera matrix $\mathbf{K}$ and correspondences $(\mathbf{u}_i, \mathbf{x}_i)$
2. Find the roots of (12), e.g., by **companion matrix** method (look it up on [www](#)!). This gives up to 4 real solutions.
3. Compute a from (6a). Compute y from (10). Compute b, c .
4. Compute point $\mathbf{T}$ from a, b, c . It leads to computing the (two) common points of three spheres centered in $\mathbf{x}_i$ with radii a, b, c . This is not as hard as it may seem, it is known as **trilateration** (one of your colleagues found it on [www](#)).

Another approach is as follows:

- (a) Find the plane π_{ab} orthogonal to $\mathbf{x}_1\mathbf{x}_2$ and passing through $\mathbf{T}$ (we leave this as an exercise). Similarly, find planes π_{ac} and π_{bc} .
 - (b) Find point $\mathbf{T}$ as the intersection of the three planes. This yields two (mirror) solutions for $\mathbf{T}$.
5. Compute λ_i from equation (4).
 6. Compute matrix $\mathbf{R}$ from equation (3) by matrix inversion. Check that this gives an orthonormal matrix, $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$. Note, we can have either $\det \mathbf{R} = +1$ or $\det \mathbf{R} = -1$.

Because of multiple solutions in steps 2 and 4b, we can get up to 16 real solutions. The correct ones of must be selected out of them using additional knowledge. This knowledge is not present in vectors $\mathbf{x}_1, \dots, \mathbf{x}_3$ and $\mathbf{v}_1, \dots, \mathbf{v}_3$ but is known from other physical properties of the camera, scene, and images. This knowledge is typically as follows:

- ◆ The signs of λ_i and $\det \mathbf{R}$ are known.
 - Note, we have not solved the P3P system (3) but a slightly different system (4). When passing from (3) to (4), we eliminated $\mathbf{R}$ and lost the signs of λ_i .
 - Correct $\mathbf{R}, \mathbf{T}$ must be compatible with the known signs of λ_i and $\det \mathbf{R}$ in equation (3).
- ◆ Other correspondences $(\mathbf{v}_i, \mathbf{x}_i)$ for $i > 3$ (either in the plane $\mathbf{x}_1\mathbf{x}_2\mathbf{x}_3$ or not) are known.
 - These correspondences must be correctly projected by equation (3).

These additional constraints should leave a single solution for $\mathbf{R}, \mathbf{T}$. If not, we must look for more additional knowledge.